

Assignment 3 Report

Task 1: Visualising Audio. I started by looking at the log-mel spectrograms for all the audio files. By looking at the graphs, it was easy to see which files were clean and which were noisy. The noisy files had bright colours spread all over the graph, which represented the background static. The clean files had clear, sharp shapes where the person was talking and dark space everywhere else.

Task 2: Filtering and Denoising To clean the noisy files, I first used a bandpass filter from the scipy library. I set the filter between 80 Hz and 7500 Hz so it would only keep the natural human voice range. Then, I performed spectral denoising from scratch using librosa. I converted the audio to frequencies, took the first half-second of the audio to figure out what the background noise looked like, and subtracted that exact noise from the rest of the clip. This successfully removed the aeroplane-like hum without hiding the human voice.

Task 3: Whisper Transcription and Tuning. I used the Whisper base model to transcribe the audio. I tested different settings like temperature, initial prompt, and beam size. I ran these tests on both the original raw audio and my new filtered audio. What I found: For the raw noisy audio, changing settings like lowering the temperature to 0 was really important to stop the model from hallucinating or making up words from the static. However, the filtered audio from Task 2 was so clean that Whisper transcribed it accurately even on the default settings.

Task 4: Voice Activity Detection. For the bonus task, I used the Silero VAD tool on the raw audio files to find the exact timestamps where a human was actually speaking. I removed the silent parts and saved the speaking parts together in a new folder. What I found: Removing the silences helped Whisper run much faster. It also completely stopped the model from guessing words during long pauses. The only small issue was that it sometimes cut off the very beginning of a word if the speaker started talking too fast.